Empirical Validation of the ReAct Paradigm for Small-Scale Research Agents

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Abstract

This study investigates the scaling of agentic capabilities in Small Language Models (SLMs). By examining the synergy between the ReAct (Reasoning and Acting) paradigm and Supervised Fine-Tuning (SFT) within an ArXiv literature retrieval task, we reveal that while Prompt Engineering provides a behavioral scaffold, SFT is essential for internalizing the underlying state-machine logic. We have open-sourced the complete dataset and reasoning trajectories on GitHub.
Here is the Repository

1 Introduction

As the field moves towards autonomous agency, achieving reliable performance in parameter-constrained environments remains a significant challenge. Small Language Models (SLMs), specifically those under 2B parameters, frequently encounter **Trajectory Collapse**, failing to maintain a coherent reasoning chain during multi-step tool interactions. This report explores whether the **ReAct (Reasoning and Acting) paradigm** can effectively mitigate these limitations in SLMs. We focus on the intrinsic effectiveness of the ReAct framework in providing a structural scaffold for Qwen3-1.7B-Instruct. By combining targeted Supervised Fine-Tuning (SFT) with the ReAct loop, we investigate the model's ability to internalize state-machine logic and maintain epistemic honesty within the specialized domain of ArXiv literature retrieval.

2 Methodology

This section describes the technical implementation of our autonomous agent, the structural composition of our benchmark dataset, and the multi-dimensional evaluation rubric used to categorize agent behavior.

2.1 Task Definition and Environment

The task involves an autonomous academic assistant retrieving hardware-related research papers from the ArXiv pre-print repository. The model is integrated with a `search_arxiv` tool, which requires the generation of three parameters: a search *query*, a *start_date*, and an *end_date*. This tool returns the 6 most relevant papers within the given time range, including titles, publication dates and abstracts. The agent must navigate the environment through a multi-turn reasoning loop: `Thought` → `Action` → `Observation` before ultimately providing a `Final Answer`.

A critical aspect of our methodology is the recognition that not all research queries yield results in the pre-print domain. We can divide queries into two categories: searchable and non-searchable.:

- **Searchable Queries**: Problems where high-quality, relevant papers exist within the specified timeframes. For these, the agent is expected to retrieve and summarize specific findings.
- **Unsearchable Queries**: Problems involving niche technologies or specific time windows where no relevant pre-prints exist. For these, the agent must demonstrate *Epistemic Honesty* by accurately reporting the absence of information.

2.2 Supervised Fine-Tuning (SFT)

We executed SFT on 95 expert-curated trajectories. The training objective was designed to internalize the state-machine logic of the ReAct cycle, specifically focusing on the transition from "Observation Failure" to "Query Refinement," a behavior rarely observed in vanilla small size models.

2.3 Categorical Evaluation

Rather than using holistic scoring, we adopt a categorical labeling approach to diagnose the specific failure and success modes of the model. Trajectories are classified into the following hierarchy:

1. **Success Modes**: Defined by the successful generation of a `Final Answer`. These outcomes are further stratified by the query's ground-truth searchability:
    - **Searchable Queries (S)**: Focuses on *Retrieval Recall*.
        - **S-Precision**: Correct retrieval, including cases with minor scope deviations.
        - **S-Honesty**: Failed retrieval but with a faithful report and no fabrication.

  - **S-Hallucination**: Reaching a conclusion through non-grounded internal reasoning.
- **Unsearchable Queries (U)**: Focuses on *Epistemic Calibration.*
  - **U-Precision**: The model identified relevant information overlooked in the standard answer and delivered a response with rigorous logical consistency.
  - **U-Honesty**: Correct identification of information absence (the model correctly states the data is missing).
  - **U-Hallucination**: Fabricating nonexistent results or papers.

2. **Failure Modes (F)**: Focuses on *Instructional Integrity.*
   - **F-Instructional**: Total failure to invoke tools.
   - **F-Operational**: Reasoning interruption or infinite looping.

Moreover, we categorize instances of hallucinations in the model's reasoning trajectory as follows:

- **Severe**: The model fails to follow formatting instructions or produce the Final Answer.
- **Moderate**: The model fabricated information that does not exist in the Observation, proceeding with hallucinated reasoning.
- **Slight**: The model demonstrated minimal hallucinations in its reasoning trajectory and maintained rigorous logical consistency.

2.4 Oracle-Based Grading

To maintain evaluation consistency, we utilized a two-stage process:

1. **Oracle Annotation**: Each question is pre-annotated with whether it is searchable, along with a logically rigorous reasoning chain as the standard answer. By comparing the reasoning trajectory of the model under evaluation with the standard reasoning trajectory, the assessment of the model's performance can reveal its information retrieval capabilities, hallucination patterns, and ultimately categorize it as mentioned.
2. **Constrained Assessment**: We employed a high-capacity model as a judge, and restricted its verdict by injecting the pre-validated searchability as an **Immutable Constraint**. This forces the judge to evaluate the agent's *process* (e.g., whether a failure to find info was an honest report or a retrieval error) based on the absolute reality of the database, rather than the agent's own potentially flawed narrative.

2.5 Experimental Configurations

To systematically decouple the individual and synergistic effects of the ReAct paradigm and Supervised Fine-Tuning, we designed a 2×2 comparative experiment. The four configurations are defined as follows:

- **Group I: Baseline (Non-ReAct Prompt)**
  This serves as the control group. We employ the Qwen3-1.7B-Instruct base model with a naive system prompt that defines the task but provides no structural guidance or few-shot exemplars. This configuration tests the model's zero-shot capability to follow complex instructions purely through its pre-trained weights.

- **Group II: Baseline (ReAct shot Prompt)**
  Using the base model, we introduce the ReAct system prompt, which includes few-shot reasoning trajectories. This configuration evaluates the efficacy of **In-Context Learning (ICL)** alone in inducing agentic behavior in small-scale models.

- **Group III: Finetuning (Non-ReAct Prompt)**
  This group utilizes the model after undergoing SFT on our curated 95 trajectories. The evaluation is conducted using the naive system prompt from Group I. This isolates the effect of **internalized logic**, testing whether the model can maintain the ReAct state-machine behavior even without explicit structural prompting.

- **Group IV: Finetuning (ReAct shot Prompt)**
  The experimental group. We combine the fine-tuned model with the ReAct prompt (including few-shot examples). This setup aims to observe the peak cumulative performance and the synergistic effect when internalized reasoning logic is reinforced by contextual guidance.

All experiments were conducted using a low-temperature decoding strategy (Temperature = 0.05) to ensure the reproducibility of the reasoning trajectories and to minimize variance in the categorical evaluation.

3 Results and Discussion

In this section, we present a quantitative analysis of the agent's performance across four ablation groups and discuss the underlying behavioral shifts induced by SFT and ReAct prompting.

The overall performance trajectory exhibits an upward trend with the introduction of ReAct prompting and SFT.

Table 1: Detailed Categorical Distribution of Reasoning Trajectories

Configuration	**Searchable (Success)**			**Unsearchable (Success)**			**Failure**	
	Prec.[a]	**Honst.**	**Noisy**	**Prec.**	**Honst.**	**Noisy**	**Inst.**	**Frag.**
Base (Non-ReAct)	0	0	0	0	3	0	37	10
Base (ReAct)	0	5	1	0	8	2	22	12
SFT (Non-ReAct)	3	6	0	0	11	3	1	26
SFT (ReAct)	**1**	**5**	**4**	**0**	**15**	**10**	**0**	**15**

Note: [a]Prec. refers to Precision; Honst. refers to Honest behavior; Noisy refers to results with reasoning halluciantion; Inst. is Instructional Collapse; Frag. is Operational Fragility (2.2 and 2.3).

Table 2: Hallucination Distribution

Configuration	**Slight**	**Moderate**	**Severe**
Base (Non-ReAct)	3	0	47
Base (ReAct)	13	3	34
SFT (Non-ReAct)	19	4	27
SFT (ReAct)	**20**	**15**	**15**

3.1 Overall Task Performance

The experimental results (Table 1) reveal a dramatic non-linear improvement in agentic reliability. The baseline model is virtually non-functional in a zero-shot agentic setting (SR: 6.0%). The introduction of few-shot ReAct prompting increases the success rate to 32.0%, but it is the supervised fine-tuning (SFT) that catalyzes the most significant leap, reaching a peak SR of 70.0% when combined with ReAct scaffolding.

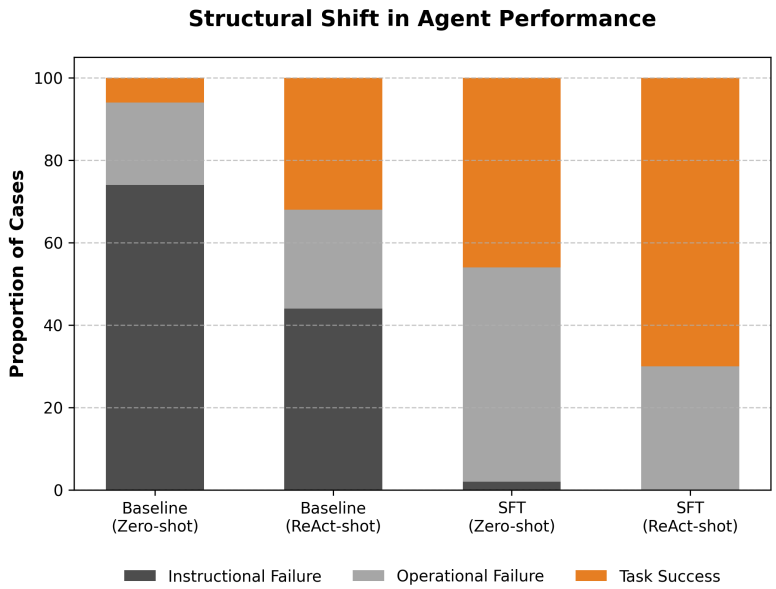

Figure 1: The Structural Shift of Failure Modes. The transition from Baseline to SFT-Full represents a migration from "Instructional Collapse" to "Operational Fragility."

3.2 The Structural Shift: From Incompetence to Fragility

As illustrated in Table 1, we observe a profound **Structural Shift** in failure modes.

- **Instructional Collapse**: In the Baseline-Zero group, 74% of cases failed because the model echoed the system prompt or outputted static text instead of tool calls. This suggests that sub-2B models lack the native inductive bias to maintain an agentic state without specialized tuning.
- **Operational Fragility**: In the SFT groups, Instructional Failure errors nearly vanish (dropped to 0% in SFT-ReAct). Instead, the primary bottleneck becomes reasoning interruption (Operational Failure). This indicates that while SFT successfully internalizes the *intent* to act, the 1.7B parameter limit manifests as a "reasoning stamina" issue, where the model loses track of the goal state after multiple observation cycles.

3.3 Calibration of Epistemic Honesty

A key highlight is the emergence of **Epistemic Honesty** in the fine-tuned agent.

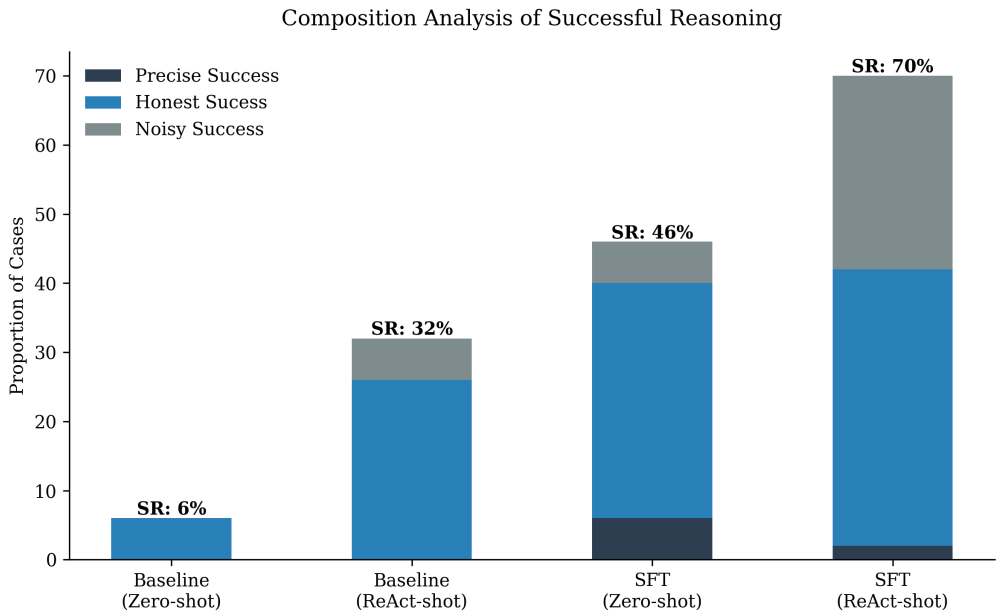

Figure 2: Success Composition Analysis.

As shown in Figure 2, the SFT-Full model does not achieve success solely through absolute precision. Rather, 40% of the total successful outcomes are categorized as **Searchable-Honest** or **Unsearchable-Honest**. In these cases, the model accurately recognized the boundaries of the ArXiv repository or its own retrieval failure and communicated this to the user. This reliability is far more valuable for real-world academic assistants than the "confident hallucinations" often observed in un-tuned models.

3.4 Reasoning Fatigue and the Signal-to-Noise Trade-off

Despite the high SR, we identify a **Reasoning Fatigue** phenomenon in the SFT-ReAct group. As reasoning chains lengthen, the likelihood of Moderate Hallucination increases, with 20% of samples falling into Category Severe Hallucination.

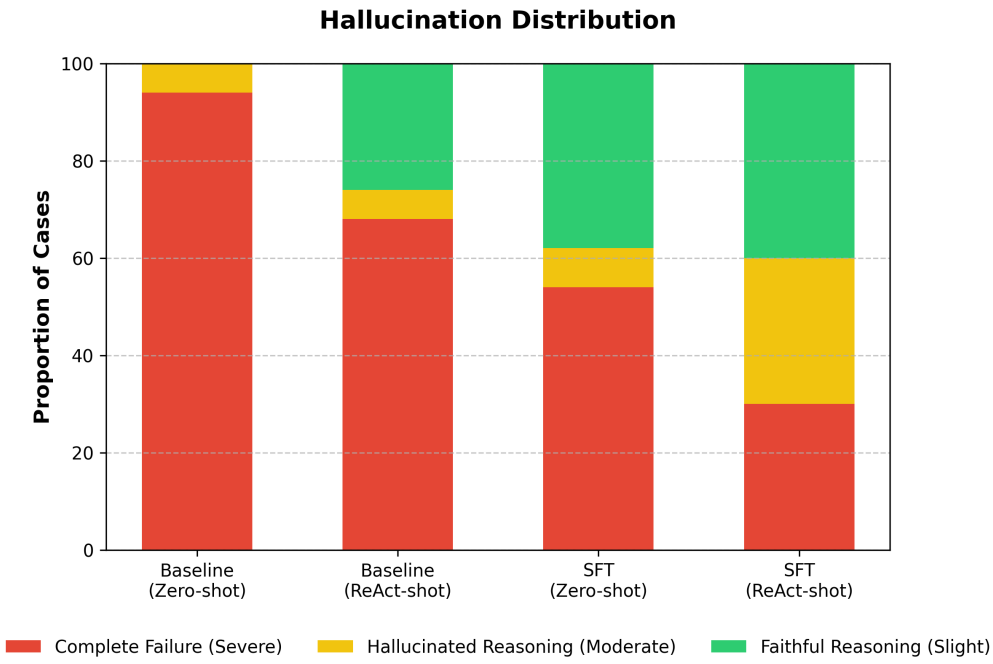

Figure 3: Distribution of Hallucination Inhibition Scores. SFT-ReAct effectively eliminates Severe Hallucinations but introduces minor reasoning noise.

The Figure 3 shows that while SFT-ReAct effectively eliminates Severe Hallucination, it shifts a significant portion of samples to Moderate Hallucination. This suggests that for SLMs, multi-turn ReAct cycles impose a heavy cognitive load. The model manages to remain on the correct logical track but begins to exhibit repetitive artifacts or minor factual inaccuracies in its internal thought process. This "Noise-to-Signal" trade-off is a fundamental constraint of 1.7B models when performing long-chain agency.

4 Conclusion

This study empirically validates the robustness and effectiveness of the ReAct paradigm in bridging the gap between small-scale model capacity and complex agentic tasks. Our findings demonstrate that the ReAct framework provides a critical structural loop—*Thought, Action, and Observation*—that enables a small size model (Qwen3-1.7B-Instruct) to navigate the dynamic ArXiv environment with high systematicity.

The results confirm that while the 1.7B model requires SFT to fully master the ReAct state-machine, it is the paradigm's inherent design that successfully orchestrates tool-use and reasoning. By adopting this framework, the agent achieved a 70% success rate and

exhibited a remarkable degree of **epistemic honesty**, proving that ReAct is a viable and powerful architecture for building reliable, specialized academic assistants even under extreme parameter constraints.